Pollution vs Progress: An Interactive Dashboard of Global Air Quality and UK Industrial Emissions

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Introduction

Air pollution remains a pressing global challenge, threatening both human health and economic well-being. Pollutants such as nitrogen oxides, ozone, carbon monoxide, and fine particulates ($PM_{2.5}$ and PM_{10}) aggravate respiratory illnesses, contribute to chronic disease, and impose substantial costs on societies. While industrial growth—across energy production, manufacturing, and transportation—has driven economic progress, it has also amplified emissions. Over recent decades, tightening environmental regulations have shifted these patterns in complex ways, and understanding both pollutant loads, and regulatory impacts is essential for crafting effective, evidence-based policies and public-health interventions.

I grew up in a city perpetually shrouded in haze, where every breath felt heavy and windows bore a persistent film. I watched friends and family struggle with asthma and bronchitis, and these experiences seeded my determination to harness data for social good. By visualizing how economic development and industrial activity intersect with air quality, I aim to equip communities and decision-makers with clear, actionable insights that bridge the gap between raw data and real-world impact.

In this project, I have developed an interactive dashboard that unites three data streams—city-level AQI readings, country-level socio-economic indicators, and UK industrial emissions—to tell a cohesive story. My goal is to provide policy makers, environmental researchers, NGOs, educators, and concerned citizens with a single platform where they can explore spatial patterns, uncover correlations, and trace temporal trends. With intuitive controls, consistent visual encodings, and guided narrative prompts, users can fluidly navigate from high-level overviews to detailed drill-downs, ultimately supporting informed decisions that balance progress with the air we breathe.

2. Design Process

In this section, I walk through how I went from a wide-ranging set of ideas to a cohesive, interactive dashboard—anchoring every choice in the theories we covered in Weeks 1–12. I revisit each of my five design sheets, explaining how principles of consistency, interaction design, visual variables, colour palettes, layout and typography, Munzner's What-Why-How framework, narrative genre, and human visual perception guided my work. Above all, I kept my audience—policy makers, environmental researchers, NGOs, educators, and the public at the forefront of every decision.

Sheet 1: Brainstorming & Ideation

My journey began with pure exploration. I sketched more than ten distinct storyboards, spanning maps, scatterplots, timelines, and hybrid combinations. Rather than fixating on one idea too soon, I grouped related concepts (for instance, choropleth vs. proportional maps, bar vs. line charts) and annotated each sketch with its **What** (the data I'd show), **Why** (the task it would support), and **How** (the visual treatment and interaction)—just as Munzner's framework and our Workshop 11 exercises recommend. In each thumbnail, I jotted down possible sliders, tooltips, and brushing interactions, and earmarked space for narrative callouts. By the end of this sheet, I had settled on five strong candidates that balanced spatial, relational, and temporal storytelling.

Sheet 2: Spatial Narrative with a Density Map

For my first full design, I asked: "How do pollutant categories distribute across countries?"

- **What–Why–How**
  - **What**: Country-level AQI categories
  - **Why**: Rank and compare pollution levels globally
  - **How**: A world density map colour-coded by AQI category, with filters for gas type, country, socio-economic factor, and year
- **Visual Variables**
  I used a single-hue, light-to-dark sequential palette (from ColorBrewer) to represent ordered AQI categories. This choice makes differences in luminance immediately clear, and avoids using shapes or textures, which are less effective for ordinal data.
- **Human Visual System**
  By leveraging pre-attentive features, the darkest countries jump out instantly (Stage 1), while Gestalt principles of proximity and continuity help viewers see regional clusters at a glance (Stage 2).
- **Interaction Design**
  I added a year slider so users can animate changes over time and tooltips on hover to reveal exact AQI values and country names. This follows Shneiderman's mantra overview first, then zoom & filter, then details-on-demand—so policy makers can spot hotspots quickly before drilling into specifics.
- **Consistency**
  I preserved this same sequential palette across all subsequent charts to reinforce perceptual compatibility and reduce cognitive load when users switch between views.

Sheet 3: Emission Composition Bubble Plot

My third sheet explores: "Which pollutants dominate emissions within a given industry at a specific moment?"

- **What–Why–How**
  - **What**: Pollutant categories (x-axis), emission magnitudes (y-axis), emission volume (bubble size), and year (hue)
  - **Why**: Compare each pollutant's share in an industry snapshot, then trace its history

- How: A bubble plot encodes position for category/value, area for magnitude, and colour for year letting viewers spot major emitters immediately and dive deeper via interaction

- **Visual Variables**
  I relied on position and area for accurate quantitative comparisons, and hue to denote time. By reusing my blue sequential palette for $PM_{2.5}$ and adding an accent hue for selections, I maintained visual coherence across sheets.

- **Human Visual System**
  Pre-attentive size and colour cues make the largest, darkest bubbles pop out first, while bubbles aligned along the x-axis group naturally by pollutant.

- **Interaction**
  I provided drop-down menus for industry and gas, plus a year slider. Hover tooltips display pollutant details and clicking a bubble links to a line chart below where users can adjust year ranges to explore temporal trends without losing context.

- **Consistency & Audience Fit**
  By reusing colour, typography, and interaction patterns (filters, tooltips, sliders) from other sheets, I ensured a seamless experience. Policy makers can pinpoint dominant gases instantly, and researchers or NGOs can drill into historical patterns for deeper analysis.

Sheet 4: Socio-Economic & AQI Composition Stacked Bar Chart

On Sheet 4 I tackle: "How do socio-economic factors and AQI categories combine to shape overall performance among the top countries?"

- **What–Why–How**
  - **What**: Multiple factors aggregated per country
  - **Why**: Compare how different metrics stack up within and between nations
  - **How**: A stacked bar chart uses bar height for totals and distinct hues for each factor segment so users can assess both aggregate scores and individual contributions

- **Visual Variables**
  I encoded total magnitude via bar height (leveraging our accuracy in judging length) and assigned each factor a unique hue echoing earlier palettes for consistency and following Bertin's ranking (hue for categorical distinction).

- **Human Visual System**
  Pre-attentive colour cues make segments pop out instantly, and stacking taps the Gestalt law of proximity to group segments by country on a shared baseline, enabling quick rank-order comparisons.

- **Interaction Design**
  Applying Shneiderman's mantra, I loaded all top N countries by default, then offered drop-down filters for AQI category and factor plus a "Top N" slider. Hover tooltips reveal exact values and percentages on demand keeping the view tidy until details are needed.
- **Consistency & Audience Fit**
  Using the same typography and interaction patterns ensures a fluid narrative flow. Policy makers can spot leading or lagging countries briefly, NGOs can isolate outliers, and educators or general users can explore how socio-economic development and air quality intertwine.

Sheet 5: Tabbed Interface for Narrative Exploration

Rather than a static grid, on Sheet 5 I organised the core views into **tabs**, so users can focus on one narrative angle spatial, compositional, relational, or temporal without cognitive overload.

- **What–Why–How**
  - **What**: Four standalone visualisations (density map; bubble plot; stacked bar; line chart) served one at a time
  - **Why**: Let users choose their path whether they want to compare regions, examine pollutant breakdowns, assess factor stacks, or trace trends
  - **How**: A tab widget toggles the active chart container; filters and tooltips remain consistent across views, so users never have to learn a new interface
- **Human Visual System**
  Presenting one chart at a time leverages figure–ground separation: the active view stands out, while others fade into the interface chrome. Consistent pre-attentive cues (hue, size) guide attention without extra cognitive load.
- **Interaction**
  Each tab follows the overview → zoom & filter → details-on-demand pattern. Default filters populate the chart on load, drop-downs and sliders re-query data instantly, and hover or click interactions work identically across tabs.
- **Consistency & Audience Fit**
  This tabbed design caters to varied user priorities—policy makers find geographic hotspots; researchers dive into correlations or stacks; educators and the public follow trends—within a single, predictable interface.

3. Implementation

3.1 Technical Implementation

I implemented the interactive dashboard as an R Shiny application, structuring my code around three core sections—data preparation, UI layout, and server logic—and drawing on examples from the official Shiny gallery and Plotly for R documentation for guidance on layout and interactivity.

Libraries & Frameworks

- **shiny** provides the reactive backbone and page structure.
- **leaflet** delivers fast, customizable world-map tiles and circle markers.
- **plotly** powers the interactive bubble, histogram, scatter, and line charts, enabling zooming, brushing, and tooltips.
- **dplyr**, **tidyr**, and **ggplot2** handle data cleaning and on-the-fly plotting when needed.
- **DT** is in place for any future table displays.
- **htmltools** and **shinyjs** let me inject custom CSS and JavaScript—for instance, to implement a dark-mode switch and tailor tooltip styles.

I based the overall structure on Shiny's "Sidebar Layout" and "NavbarPage" templates, and borrowed CSS snippets from the Shiny Dashboard ThemeSwitcher example to define consistent light/dark theme variables, typography, margins, and colour palettes across every component.

Data Preparation
A single R script ingests three CSVs—city-level AQI, country socio-economic data, and UK emissions—using read.csv() and dplyr::left_join(). I trimmed whitespace, converted year and emission fields to numeric, and dropped missing entries. Tidying the UK emissions into a long format and computing per-industry maximums for the bubble plot was particularly intensive, requiring me to strike a balance between clear, maintainable code and efficient reactive performance.

Deviations from the Original Design
Originally, I planned four distinct tabs (Map; Composition; Stack; Trends). Early user feedback showed a preference for grouping all global-AQI views under one tab, so I merged the pollutant-bubble and socio-economic scatter into a single "Global AQI" tab. This streamlined navigation and reduced UI complexity without sacrificing functionality.

Key Challenges

1. **Reactive Dependencies**: Tuning filters, sliders, and select inputs so they update displays efficiently—without unnecessarily reloading large data frames—was critical.
2. **Plotly Customization**: Adjusting bubble jitter and fine-tuning sizeref, sizemin, and sizemax took several iterations to maintain clarity, especially for outliers like benzene.
3. **Styling & Theming**: Defining CSS variables for both light and dark modes and ensuring they loaded correctly across different browsers involved detailed debugging.

By combining Shiny’s reactive model with Leaflet and Plotly, I delivered a flexible, high-performance dashboard that closely follows my design rationale while meeting practical requirements around usability and speed.

3.2 Interactive Narrative Visualisation Implementation

I brought my five design-sheet concepts together in a single Shiny dashboard that weaves two complementary stories—one about global air quality, the other about UK industrial emissions. By combining interactive charts, consistent styling, and on-screen guidance, I’ve created an environment where policy makers, researchers, NGOs, educators, and the public can uncover insights through exploration.

3.2.1 Overall Layout and Narrative Flow

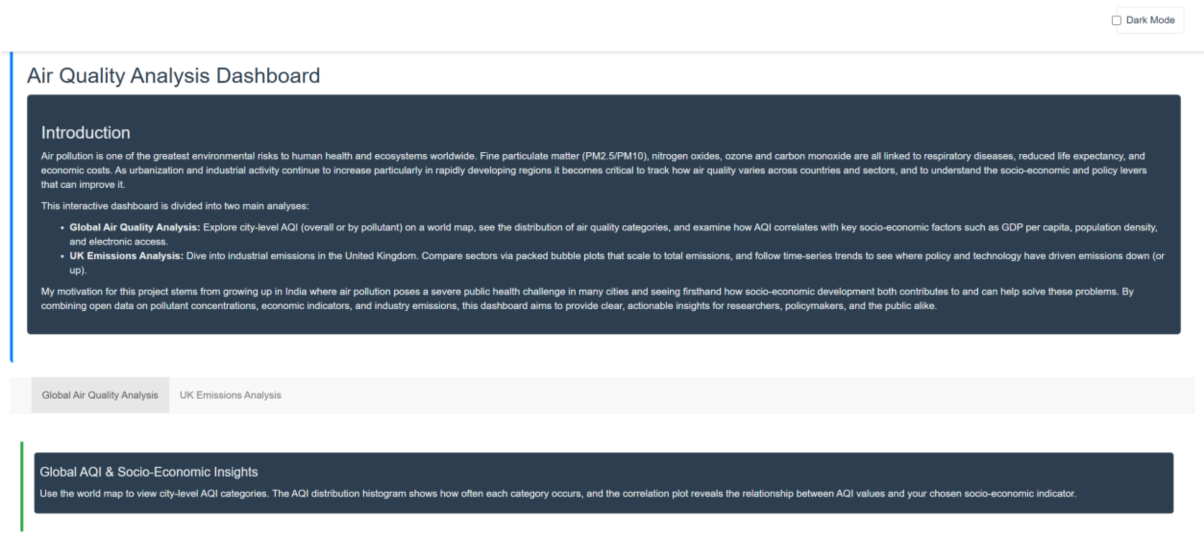
As soon as the app opens, users are greeted by a concise **Introduction** panel (Figure 1) that orients them to the dashboard’s scope:

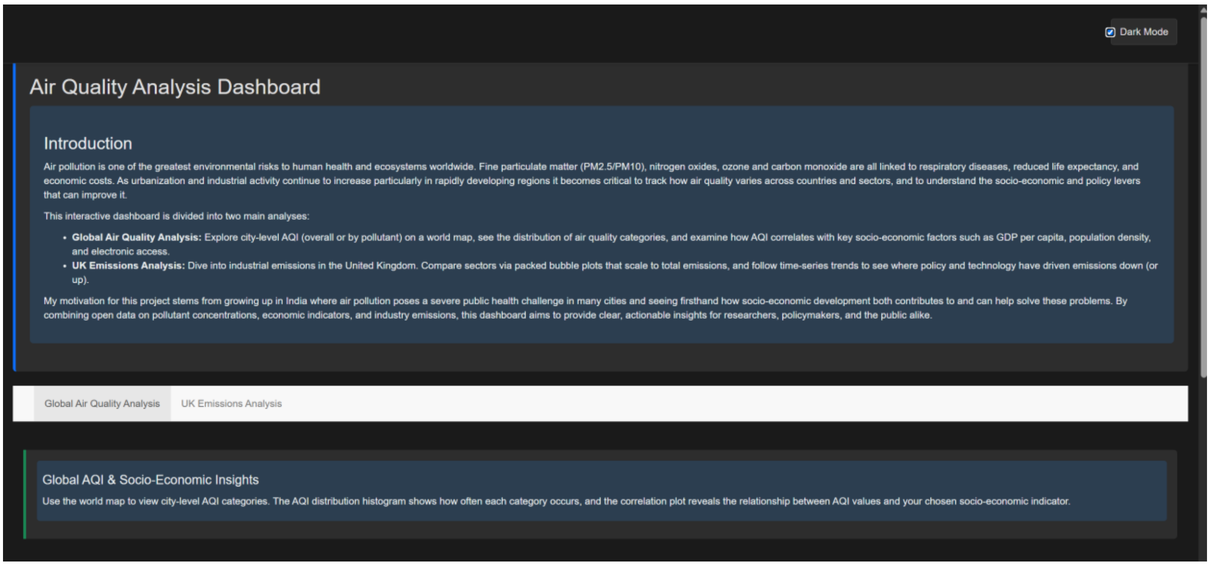
- **Context**: the health and economic consequences of air pollution worldwide and in the UK
- **Scope**: two main analysis tracks (Global AQI vs. UK Emissions)
- **Controls**: top-level tabs for “Global Air Quality Analysis” and “UK Emissions Analysis,” plus a persistent dark-mode switch

This landing view works like the title slide of a presentation, framing our guiding question—“How does air quality vary across countries and industries?”—and inviting users to dive in. Beneath it, a **navbar** holds the two narrative tabs. Each tab combines:

1. A coloured **tab-description** box that summarizes the story segment (“Global AQI & Socio-Economic Insights” or “UK Industrial Emissions Overview”).
2. A sidebar of filters and sliders tailored to that analysis.
3. A main panel where the charts appear, framed by italicized captions that remind users how to interact.

By re-using the same CSS variables, typography, and colour palette across both tabs, I deliver a seamless visual progression, reinforcing that these are facets of one unified investigation.





3.2.2 Global Air Quality Analysis

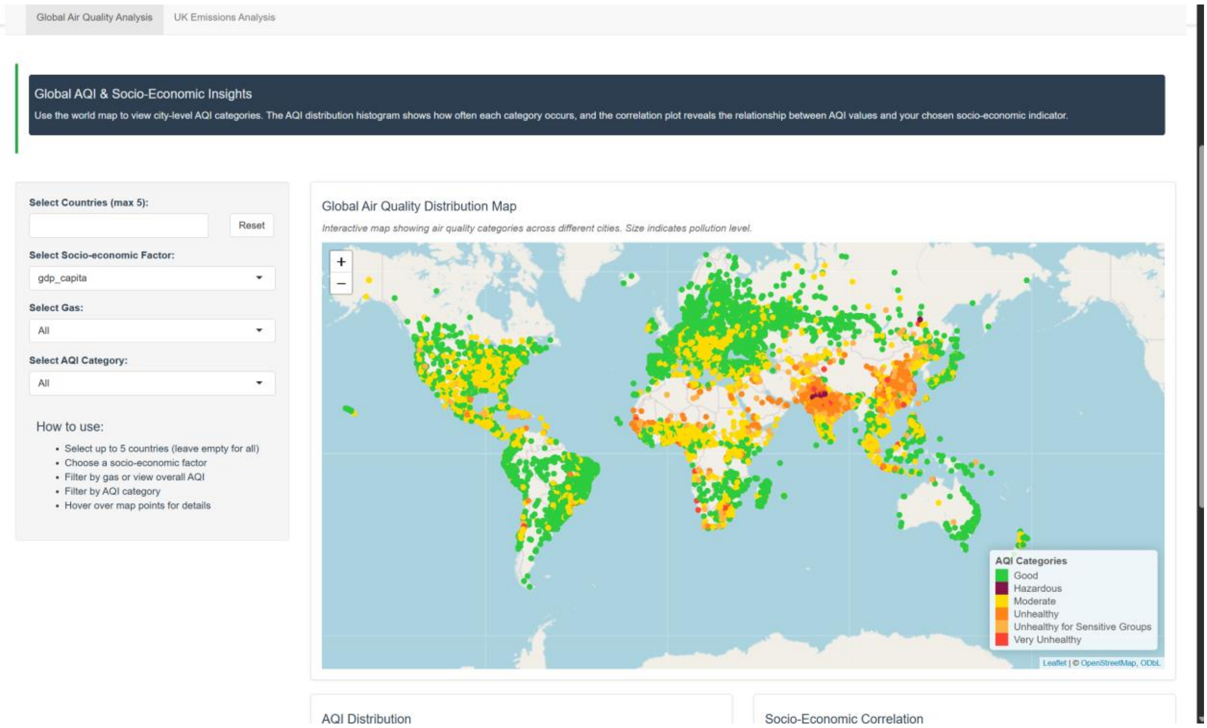
In the **Global Air Quality Analysis** tab (Figure 2), I link three views so users can explore spatial, categorical, and relational dimensions in concert:

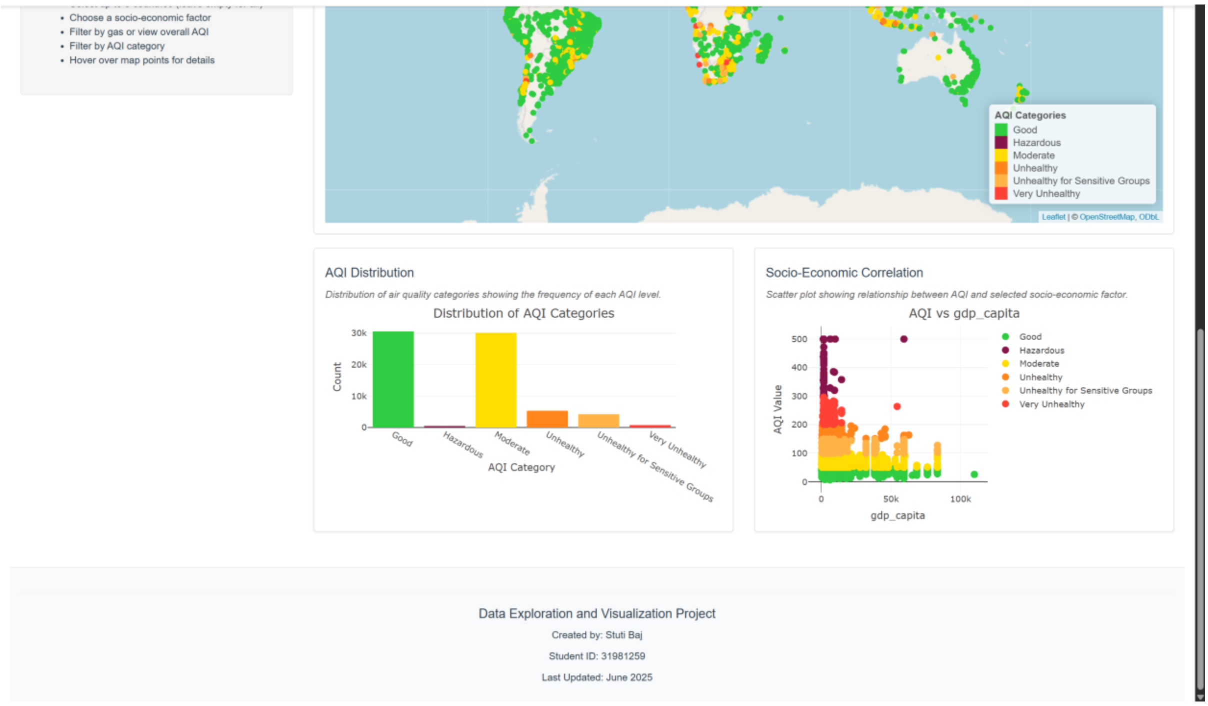
1. **Interactive World Map (Leaflet)**
   - **Story**: "Discover where city-level AQI categories cluster around the globe."
   - **Controls**: multi-select up to five countries, dropdowns for AQI category, pollutant type, and socio-economic factor
   - **Interaction**: hovering over any marker reveals city name, AQI value and category, plus the chosen socio-economic metric
2. **AQI Distribution Histogram (Plotly)**
   - **Story**: "Which AQI categories occur most often?"

- **Interaction**: this histogram responds to the same filters as the map—if you isolate “Unhealthy” and “Ozone,” the bars redraw to reflect only those records

3. **AQI vs. Socio-Economic Scatterplot (Plotly)**
    - **Story**: “How does air quality correlate with your selected economic or social indicator?”
    - **Interaction**: points are coloured by AQI category; hovering shows exact values. In a planned enhancement, brushing or clicking a point will highlight that city on the map.

Together, these coordinated views implement Shneiderman’s mantra—overview, zoom & filter, then details-on-demand—so users can move fluidly from big-picture patterns to precise data.




• Choose a socio-economic factor
• Filter by gas or view overall AQI
• Filter by AQI category
• Hover over map points for details
AQI Categories
Good
Hazardous
Moderate
Unhealthy
Unhealthy for Sensitive Groups
Very Unhealthy
Leaflet | © OpenStreetMap, ODbL
AQI Distribution
Distribution of air quality categories showing the frequency of each AQI level.
Distribution of AQI Categories
Count
30k
20k
10k
0
Good
Hazardous
Moderate
Unhealthy
Unhealthy for Sensitive Groups
Very Unhealthy
AQI Category
Socio-Economic Correlation
Scatter plot showing relationship between AQI and selected socio-economic factor.
AQI vs gdp_capita
AQI Value
500
400
300
200
100
0
50k
100k
gdp_capita
Good
Hazardous
Moderate
Unhealthy
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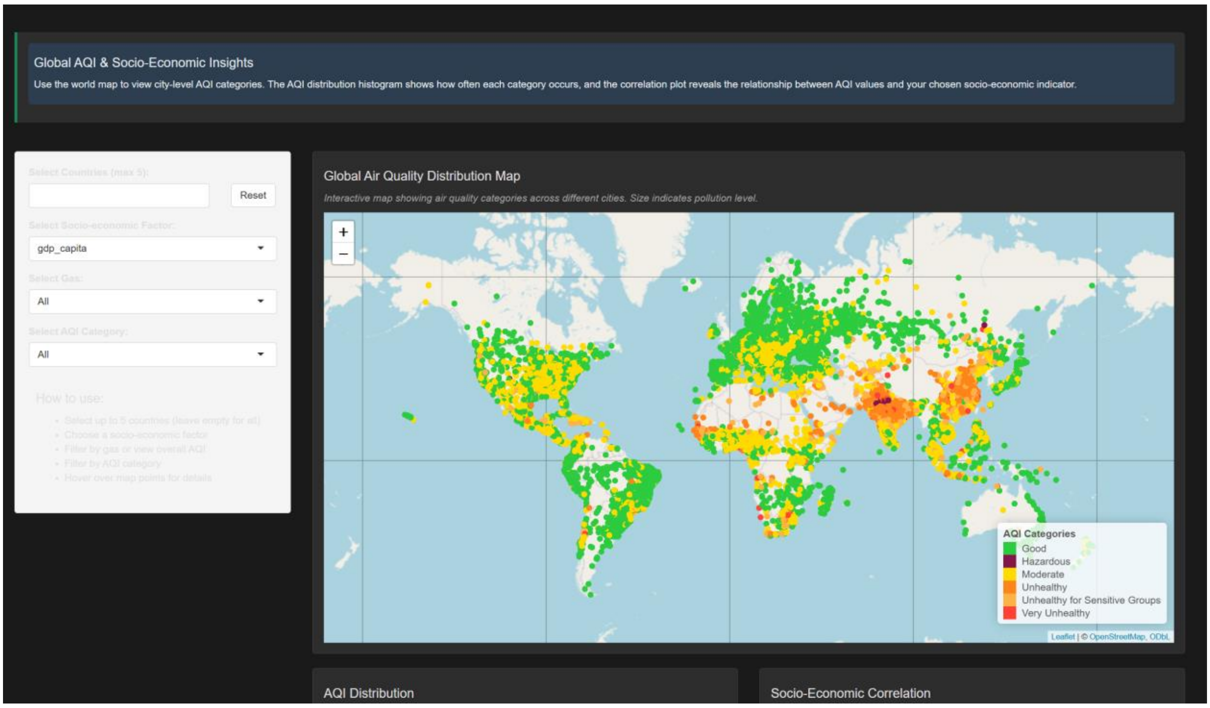
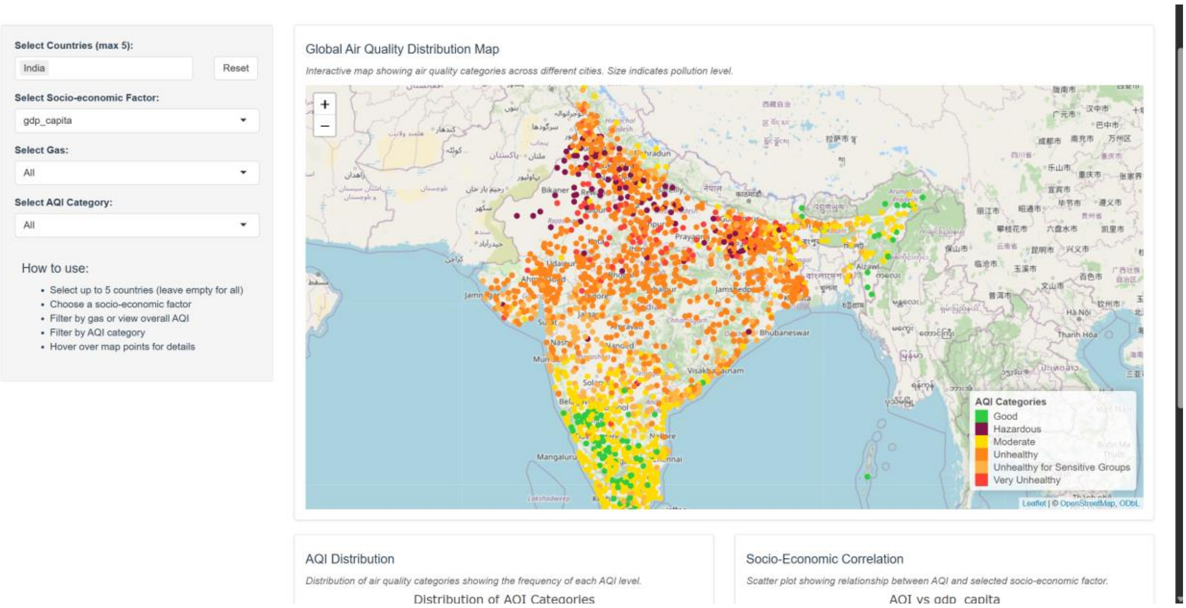
Created by: Stuti Baj
Student ID: 31981259
Last Updated: June 2025


Global AQI & Socio-Economic Insights
Use the world map to view city-level AQI categories. The AQI distribution histogram shows how often each category occurs, and the correlation plot reveals the relationship between AQI values and your chosen socio-economic indicator.
Reset
gdp_capita
All
All
Global Air Quality Distribution Map
Interactive map showing air quality categories across different cities. Size indicates pollution level.
AQI Categories
Good
Hazardous
Moderate
Unhealthy
Unhealthy for Sensitive Groups
Very Unhealthy
Leaflet | © OpenStreetMap, ODbL
AQI Distribution
Socio-Economic Correlation


Select Countries (max 5):
India
Reset
Select Socio-economic Factor:
gdp_capita
Select Gas:
All
Select AQI Category:
All
How to use:
Select up to 5 countries (leave empty for all)
Choose a socio-economic factor
Filter by gas or view overall AQI
Filter by AQI category
Hover over map points for details
Global Air Quality Distribution Map
Interactive map showing air quality categories across different cities. Size indicates pollution level.
AQI Categories
Good
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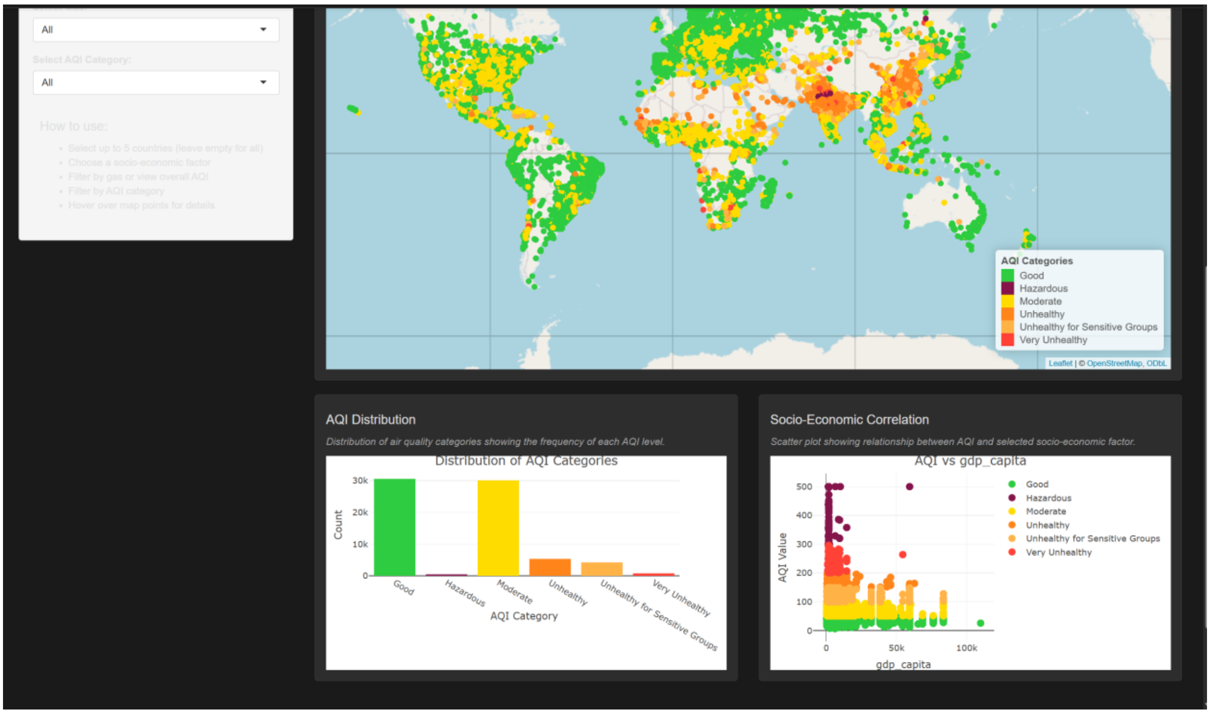
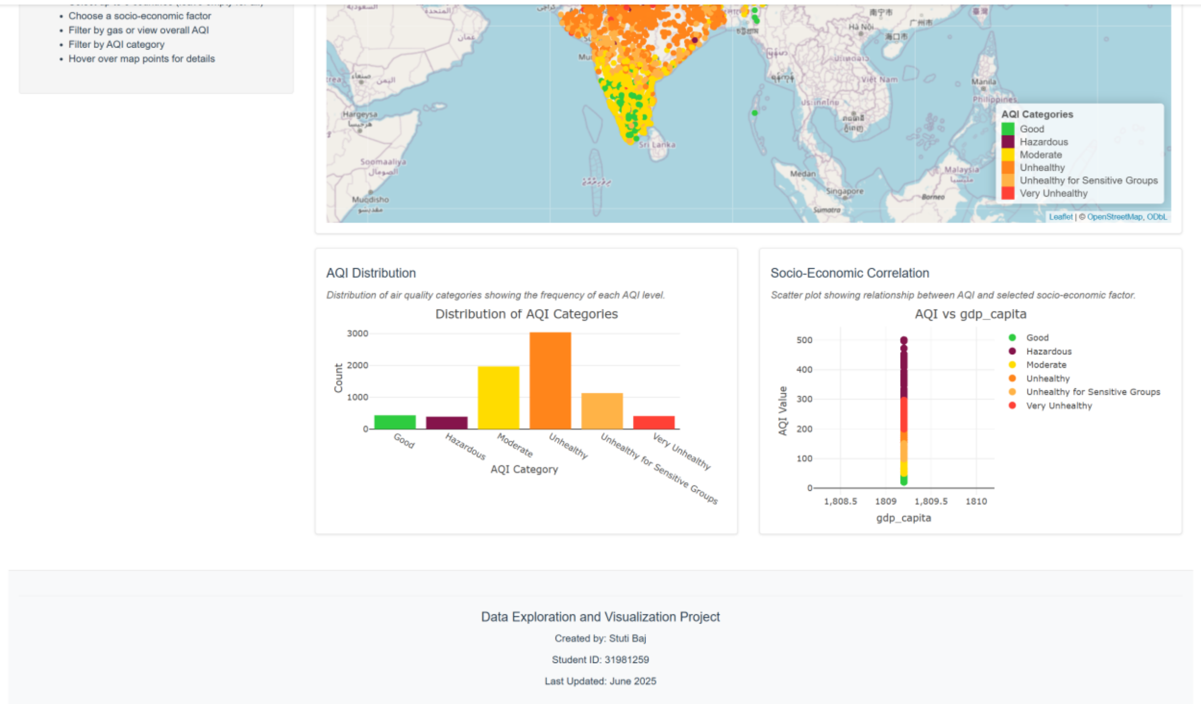
Unhealthy
Unhealthy for Sensitive Groups
Very Unhealthy
Leaflet | © OpenStreetMap, ODbL
AQI Distribution
Distribution of air quality categories showing the frequency of each AQI level.
Distribution of AQI Categories
Socio-Economic Correlation
Scatter plot showing relationship between AQI and selected socio-economic factor.
AQI vs gdp_capita


All
Select AQI Category:
All
How to use:
AQI Categories
Good
Hazardous
Moderate
Unhealthy
Unhealthy for Sensitive Groups
Very Unhealthy
Leaflet | © OpenStreetMap, ODbL
AQI Distribution
Distribution of air quality categories showing the frequency of each AQI level.
Distribution of AQI Categories
Count
30k
20k
10k
0
Good
Hazardous
Moderate
Unhealthy
Unhealthy for Sensitive Groups
Very Unhealthy
AQI Category
Socio-Economic Correlation
Scatter plot showing relationship between AQI and selected socio-economic factor.
AQI vs gdp_capita
AQI Value
500
400
300
200
100
0
0
50k
100k
gdp_capita
Good
Hazardous
Moderate
Unhealthy
Unhealthy for Sensitive Groups
Very Unhealthy

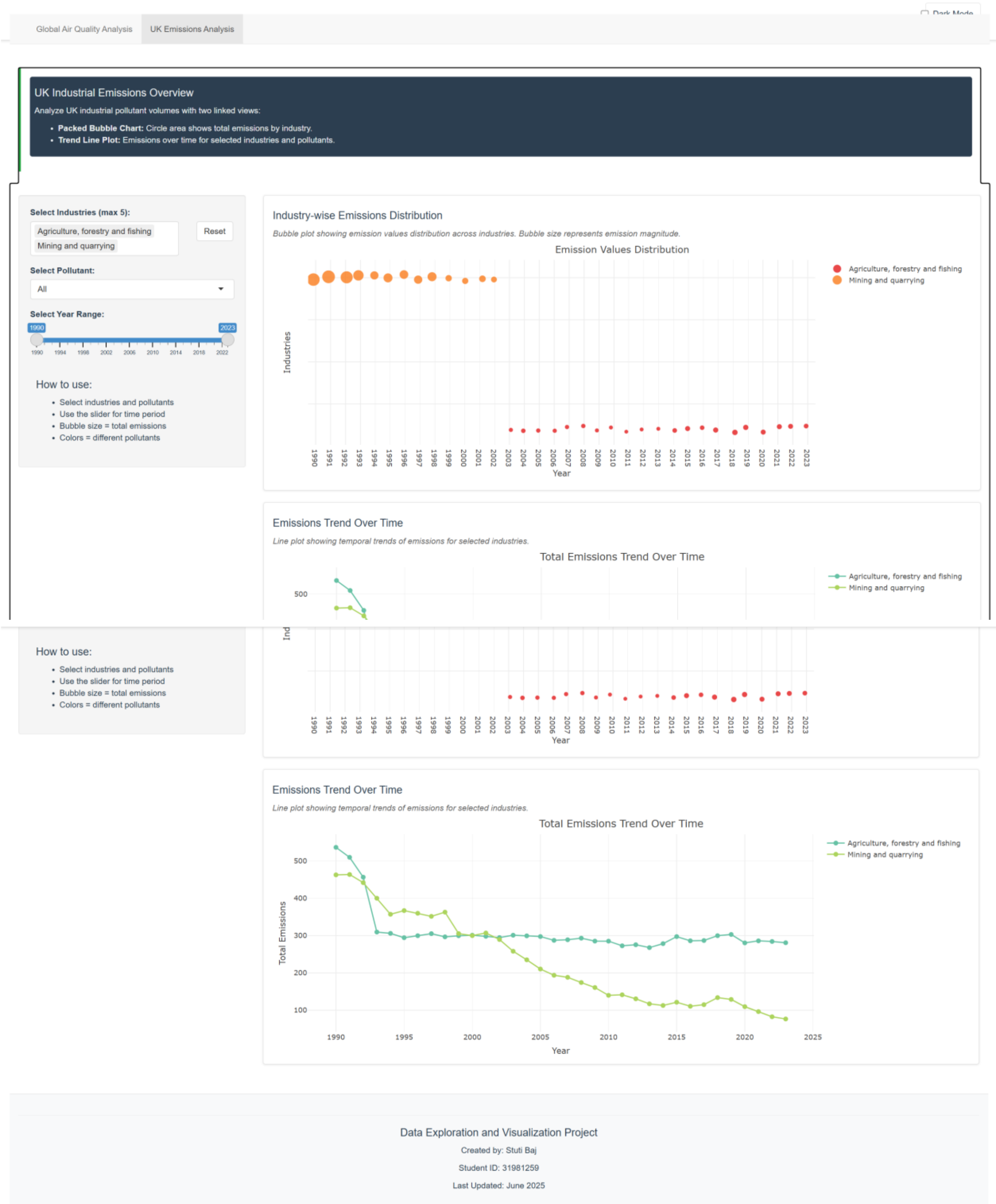


3.2.3 UK Emissions Analysis

Switching to **UK Emissions Analysis** (Figure 3) reveals a two-stage exploration of industrial pollutants:

1. **Packed Bubble Chart (Plotly)**
   - **Story**: "Which industries release the most emissions over your selected timeframe?"
   - **Controls**: multi-select up to five industries, pollutant dropdown, and year-range slider
   - **Interaction**: bubble area encodes emission volume; colour denotes pollutant type. Hover tooltips show year, industry, pollutant, and emission value (with a log-reverse for benzene where appropriate).
2. **Emissions Trend Line Chart (Plotly)**
   - **Story**: "How have emissions in your chosen industries evolved over time?"
   - **Interaction**: once industries and pollutant are selected, the line chart updates to plot annual totals per industry. Hovering over a line reveals exact values, and clicking legend entries toggles individual traces on or off.

This two-step narrative lets users compare industries side by side, then trace their progress (or setbacks) over the years.


Global Air Quality Analysis
UK Emissions Analysis
Dark Mode
UK Industrial Emissions Overview
Analyze UK industrial pollutant volumes with two linked views:
Packed Bubble Chart: Circle area shows total emissions by industry.
Trend Line Plot: Emissions over time for selected industries and pollutants.
Select Industries (max 5):
Agriculture, forestry and fishing
Mining and quarrying
Reset
Select Pollutant:
All
Select Year Range:
1990
2023
How to use:
Select industries and pollutants
Use the slider for time period
Bubble size = total emissions
Colors = different pollutants
Industry-wise Emissions Distribution
Bubble plot showing emission values distribution across industries. Bubble size represents emission magnitude.
Emission Values Distribution
Industries
Year
Agriculture, forestry and fishing
Mining and quarrying
Emissions Trend Over Time
Line plot showing temporal trends of emissions for selected industries.
Total Emissions Trend Over Time
Total Emissions
Year
Data Exploration and Visualization Project
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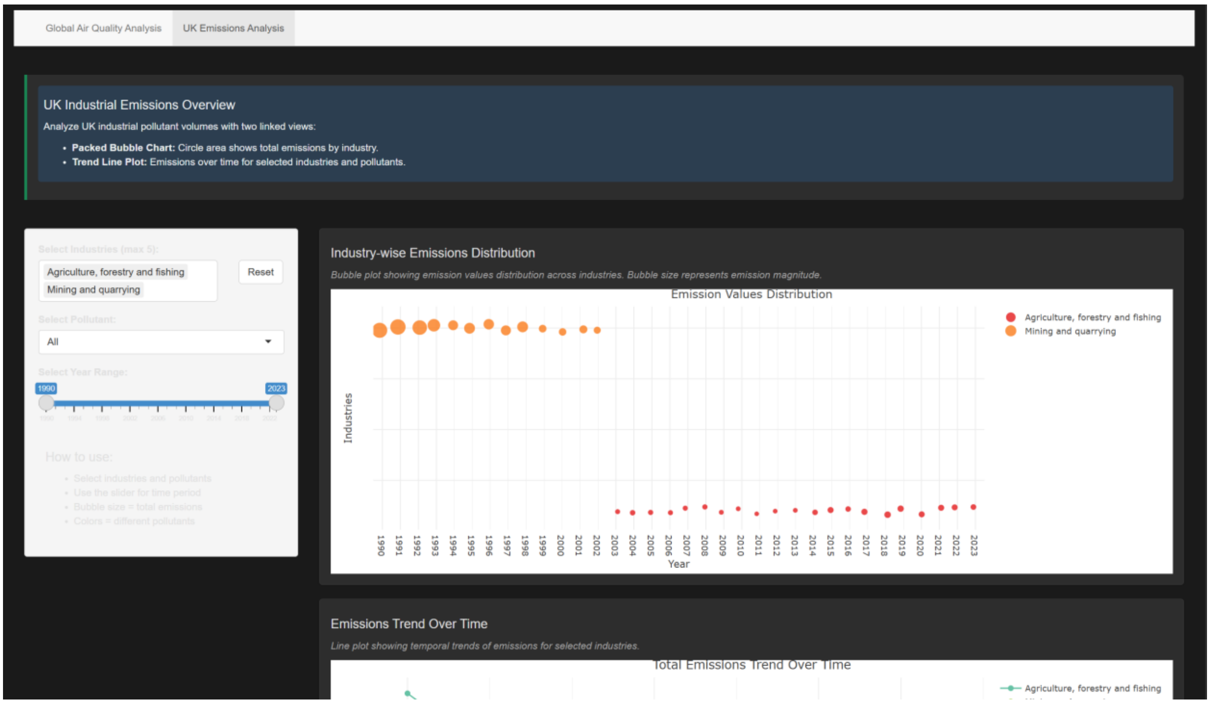
Mining and quarrying
Select Pollutant:
All
Select Year Range:
1990
2023
How to use:
Select industries and pollutants
Use the slider for time period
Bubble size = total emissions
Colors = different pollutants
Emission Values Distribution
Agriculture, forestry and fishing
Mining and quarrying
Industries
1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023
Year
Emissions Trend Over Time
Line plot showing temporal trends of emissions for selected industries.
Total Emissions Trend Over Time
Agriculture, forestry and fishing
Mining and quarrying
Total Emissions
500
400
300
200
100
1990 1995 2000 2005 2010 2015 2020 2025
Year


Global Air Quality Analysis
UK Emissions Analysis
UK Industrial Emissions Overview
Analyze UK industrial pollutant volumes with two linked views:
Packed Bubble Chart: Circle area shows total emissions by industry.
Trend Line Plot: Emissions over time for selected industries and pollutants.
Select Industries (max 5):
Agriculture, forestry and fishing
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Reset
Select Pollutant:
All
Select Year Range:
1990
2023
How to use:
Select industries and pollutants
Use the slider for time period
Bubble size = total emissions
Colors = different pollutants
Industry-wise Emissions Distribution
Bubble plot showing emission values distribution across industries. Bubble size represents emission magnitude.
Emission Values Distribution
Agriculture, forestry and fishing
Mining and quarrying
Industries
1990 1991 1992 1993 1994 1995 1996 1997 1998 1999 2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018 2019 2020 2021 2022 2023
Year
Emissions Trend Over Time
Line plot showing temporal trends of emissions for selected industries.
Total Emissions Trend Over Time
Agriculture, forestry and fishing

3.2.4 Narrative Techniques and Audience Engagement

To guide interpretation and keep cognitive load manageable, I applied several storytelling strategies:

- **Contextual Prompts**: Each tab begins with a narrative summary box, framing the upcoming chart as a self-contained story.

- **Progressive Disclosure**: Only essential controls and charts appear initially; additional details (tooltips, exact numbers) surface on demand.
- **Hybrid Martini-Glass Structure**: I lead with an author-driven overview (map or bubble chart), then open the interface for reader-driven exploration (histograms, scatterplots, line charts).
- **Consistent Visual Variables**: Repeating the same sequential blue palette for $PM_{2.5}$ and consistent accent hues ensures users carry familiar colour associations across views.
- **Accessibility & Usability**: A dark-mode toggle, clear typography, generous spacing, and colourblind-safe palettes make the dashboard approachable for diverse audiences.

By combining these techniques, I've created an interactive narrative that feels both guided and open-ended—an environment where users can effortlessly transition from big-picture insights to detailed drill-downs.

3.3 Using the Implementation

Global Air Quality Analysis Tab

1. **Sidebar Controls**
    - **Country Selector**: Pick up to five countries,(adding a sixth automatically removes the oldest choice). Use the **Reset** button to clear all selections.
    - **Socio-economic Factor**: Choose one metric to plot against AQI.
    - **Pollutant & AQI Category Filters**: Narrow the data to a specific gas type or AQI band.
2. **Interactive World Map**
    - **Hover Tooltips**: Mousing over a circle reveals the city name, its AQI category and value, plus the selected socio-economic indicator.
3. **AQI Distribution & Correlation Charts**
    - Both plots automatically refresh whenever you change any sidebar setting.
    - Clicking a bar in the histogram or a point in the scatterplot will highlight that same city on the map, linking details on demand.

UK Emissions Analysis Tab

1. **Sidebar Controls**
    - **Industry Selector**: Choose up to five sectors; adding more will drop the earliest selection. Click **Reset** to clear all.

- **Pollutant & Year-Range Slider**: Filter the chart by pollutant type and time-period.

2. **Packed Bubble Chart**
   - **Encoding**: Bubble area reflects total emissions; bubble colour denotes pollutant.
   - **Hover Details**: Viewing a bubble displays the year, industry name, pollutant type, and exact emission value (with benzene values automatically back-transformed from the log scale).
3. **Emissions Trend Line**
   - The line chart updates to match your industry, pollutant, and year-range choices.
   - Hovering over any point shows detailed annual values, and clicking entries in the legend toggles each industry's line on or off.

Conclusion

This project has shown how a thoughtfully designed narrative visualisation can turn intricate environmental datasets into intuitive, actionable stories for a wide range of users. I delivered a unified Shiny dashboard that steers audiences—from decision makers to concerned citizens—through three interlinked narratives: mapping global air-quality patterns, uncovering links between pollution and socio-economic factors, and tracing industry emissions over time. By working through five design sheets, I honed my selection of visual encodings, interactive features, and storytelling techniques so that each view stands on its own while also flowing seamlessly into the next.

I came to appreciate the value of rooting every design choice in established theory—Munzner's What-Why-How framework, Bertin's hierarchy of visual variables, and human-vision principles. Translating these ideas into code surfaced real-world challenges too: optimizing Shiny's reactive data flows, fine-tuning Plotly's bubble sizing, and balancing styling consistency with browser performance. These experiences deepened my understanding of the trade-offs between an ideal design and practical constraints.

Looking back, I see opportunities to make the narrative even more self-explanatory—by embedding guided annotations and storytelling cues directly onto the charts, particularly in the UK Emissions view, so users rely less on sidebar instructions. Adding an optional tutorial overlay could also help newcomers master features like brushing and drill-down linking. In future iterations, I plan to incorporate predictive models to forecast air-quality under different policy scenarios and to enable users to annotate or export data for further analysis.

Overall, this project sharpened my skills in theory-driven design, interactive development, and iterative refinement. I'm excited to build on this foundation and create even more engaging, insightful narrative visualisations in the future.

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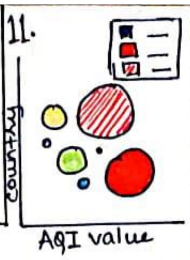
Acknowledgement of AI

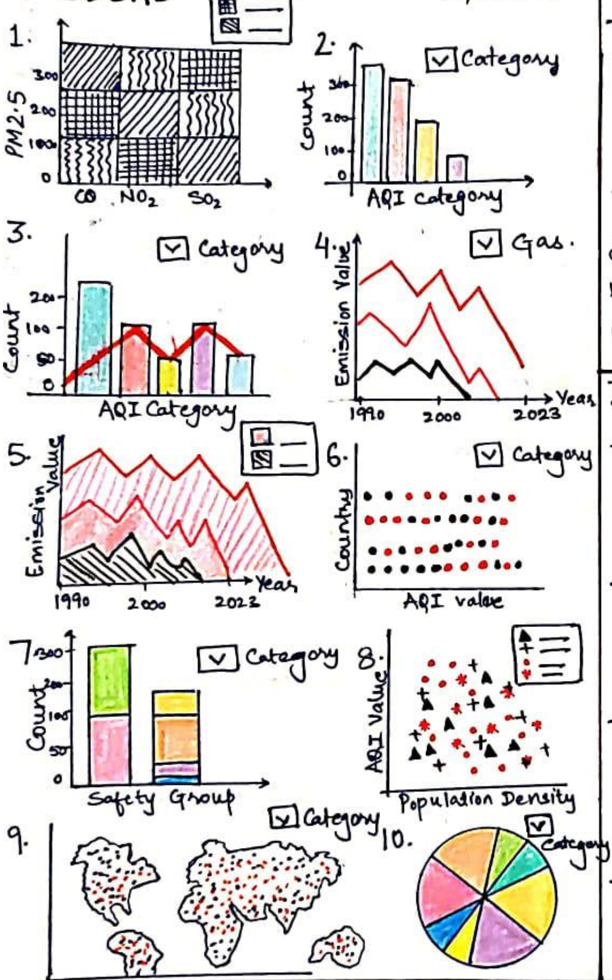
I acknowledge the use of generative AI (ChatGPT) (https://chatgpt.com) to assist in the structure, formatting, strategy, and proofreading of this assignment. The final content and analysis are my own original work, with all referenced materials accurately cited in accordance with Monash University's APA guidelines.

TITLE: Pollution Vs Progress.
AUTHOR: Stuti Bajaj
DATE: 18-05-2025
SHEET: SHEET-1
TASK: Brainstorm.

11.



1. IDEAS



2. FILTER

- Chart 1: Irrelevant as only few input features can be selected instead of large matrix of dimensions. Might not be able to communicate the message clearly and might overwhelm the users without providing meaningful comparision.

- Chart 10: Not suitable for comparing multiple pollutants or factors influencing AQI. Ineffective when dealing with numerous features, variables. It lacks clarity.

- Additional features that can be added:-

a) Filtering by AQI category
b) Filtering by pollutant type
c) Filtering by PM2.5 as it aligns with WHO guidelines

3. CATEGORIZE

— 2,3,7: It can be useful for comparing AQI category accross countries.

— 4,5,: Best for understanding trends of gases and their emission values

— 6,8,11: It can be helpful in understanding relationships of AQI v/s. Factors

— 9: Ideal for representing geographic spread of AQI.

4. COMBINE AND REFINE

— 2,3,7: can be combined in a bar chart to represent two factors

— 11 + 4,5: can be combined for understanding the data with more insights and time trends. More colours can be used to understand about more features.

— 9: will help in understanding the position of countries in the world w.r.t AQI.

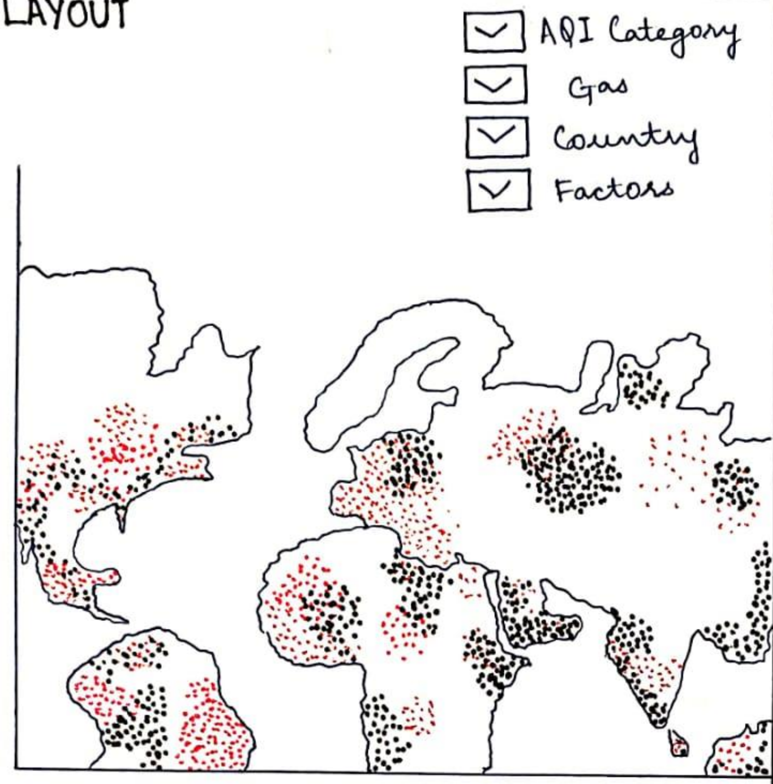
5. QUESTION

- Will showing the visualisation using map help users to understand the global air quality changes?

- Can filtering by pollutant type, category, region help users understand and explore specific problems

- Will showing a focused example like UK trends help user to understand industrial sector influence to AQI

LAYOUT

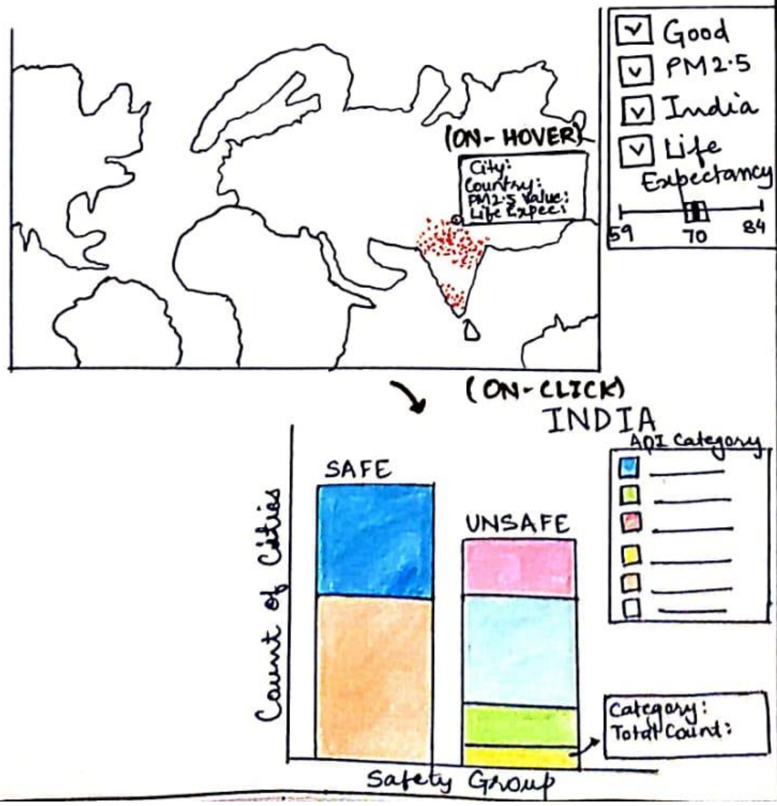


TITLE: DVP: Design
AUTHOR: Stuti Bai
DATE: 18-05-2025
SHEET: 2
TASK: Initial Design 1

OPERATIONS:

- 4 drop down menu for user to choose and customize the data that they desire.
- Hovering mouse over the dot will display values.
- Stacked bar chart allows better understanding of different distribution of various AQI category in the country
- Sliders used for factors gives a better visualisation of data

FOCUS



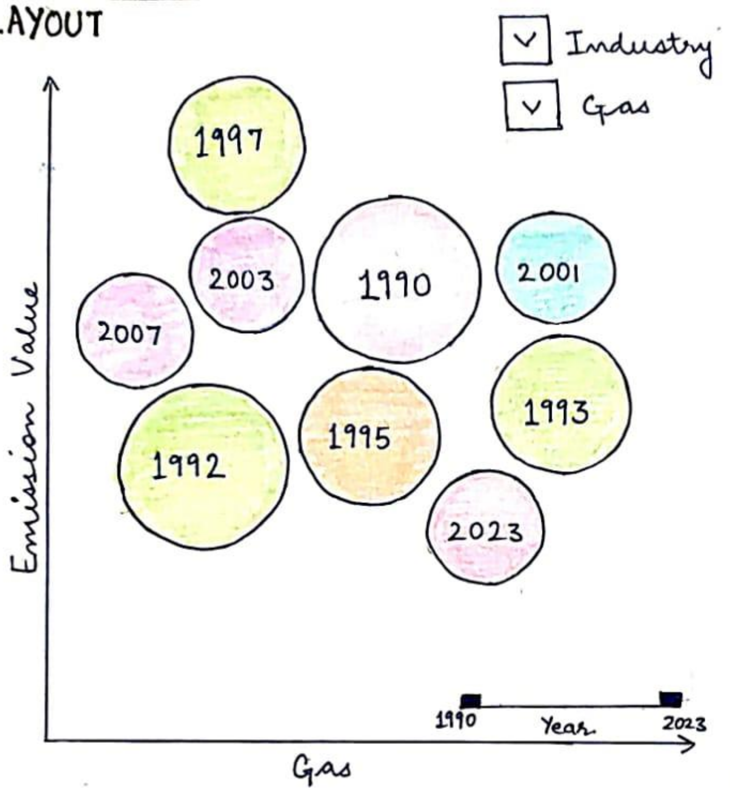
DISCUSSION

+ve : • Filters and drop-down menu allow better understanding of the visualisation

• Stacked bar chart helps in understanding distribution of AQI categories and safety groups

-ve : If users select multiple factors or countries the visualisation can become complicated and may not be able to provide better understanding of data.

LAYOUT

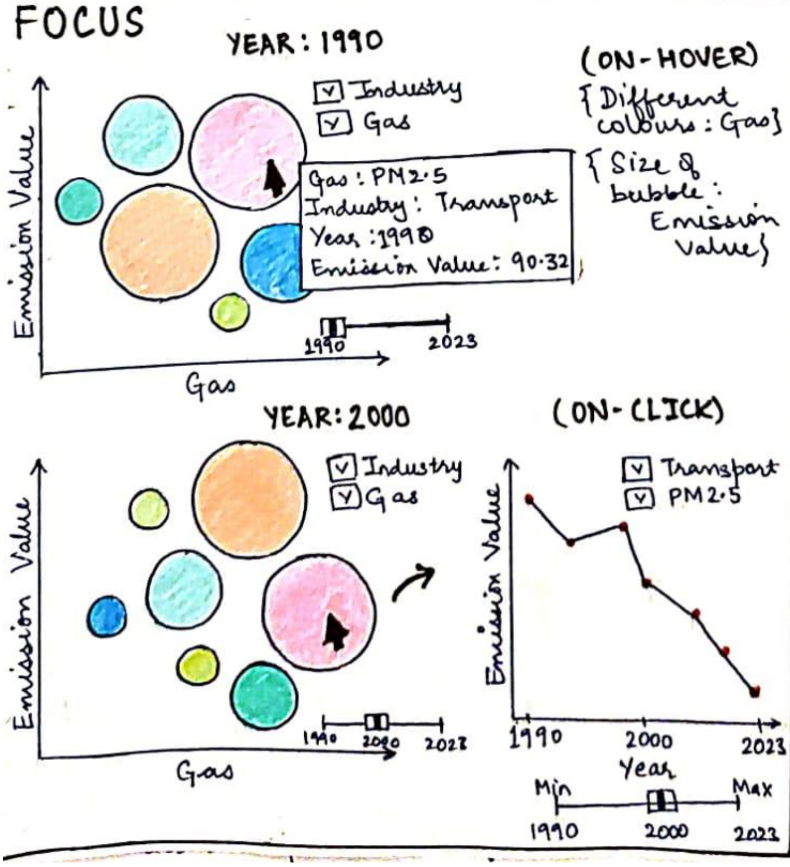


TITLE: DVP : Design
AUTHOR: Stuti Baj
DATE: 18-05-2025
SHEET: 3
TASK: Intial Design 2

OPERATIONS

- Two drop down menu to filter category of industry and gas. It is for user to customize the data that they desire to know.
- Hovering the mouse over bubble will display the gas, industry, year and its emission value
- On clicking a bubble it leads to a line graph displaying changes over time. A slider is added to change min or max of the year. to understand the changes over time

FOCUS



DISCUSSION

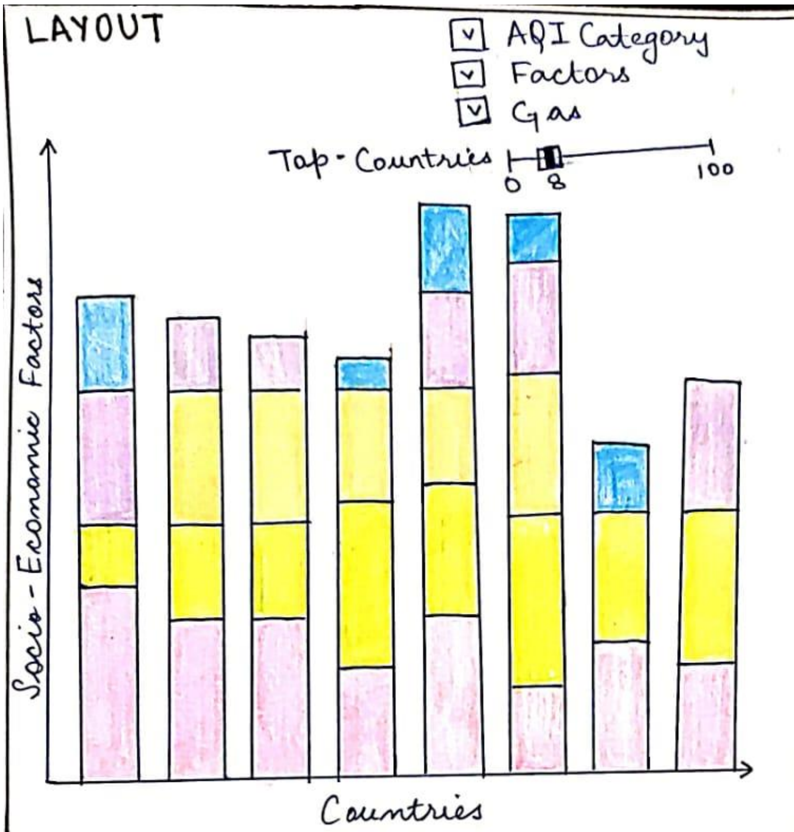
+ve: Filters such as drop down menu, slider gives a better understanding of data along with line chart to display series of information.

-ve: • It is hard to understand unless the bubble is clicked on. and you get a look of the line chart.

• It is only visualising about Uk and not about the world, it doesn't address the issue of air quality globally.

TITLE: DVP : Design
AUTHOR: Stuti Baj
DATE: 18 - 05 - 2025
SHEET: 4
TASK: Initial Design 3

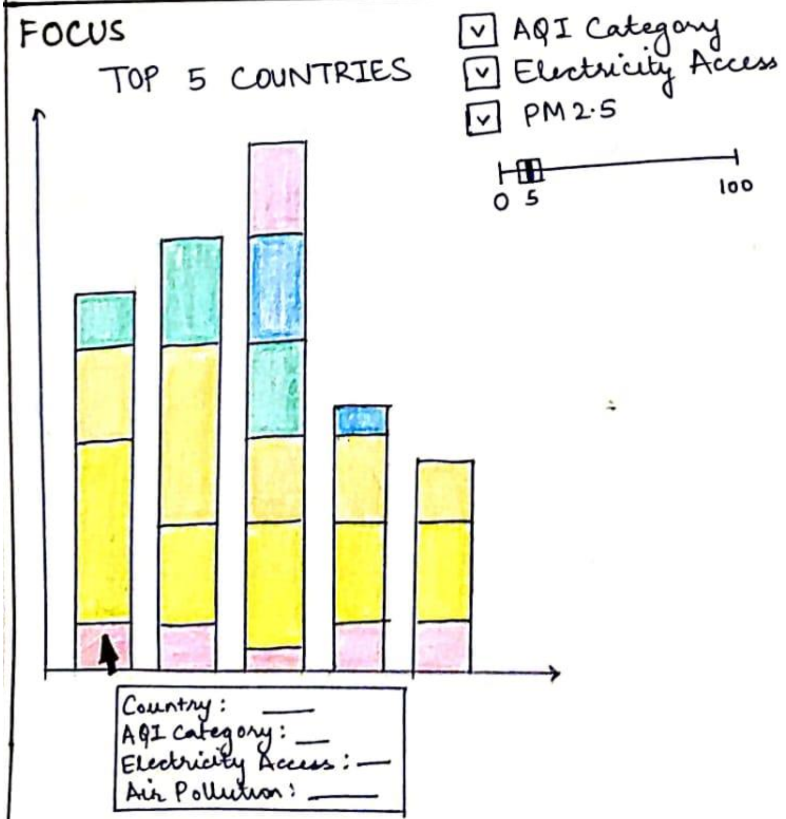
LAYOUT



OPERATIONS

- 3 drop down menus to filter the users desired categories for better visualisation
- A slider to choose top 'n' countries to make comparisions
- On hover we can see the display that provides a clear picture of the values.

FOCUS



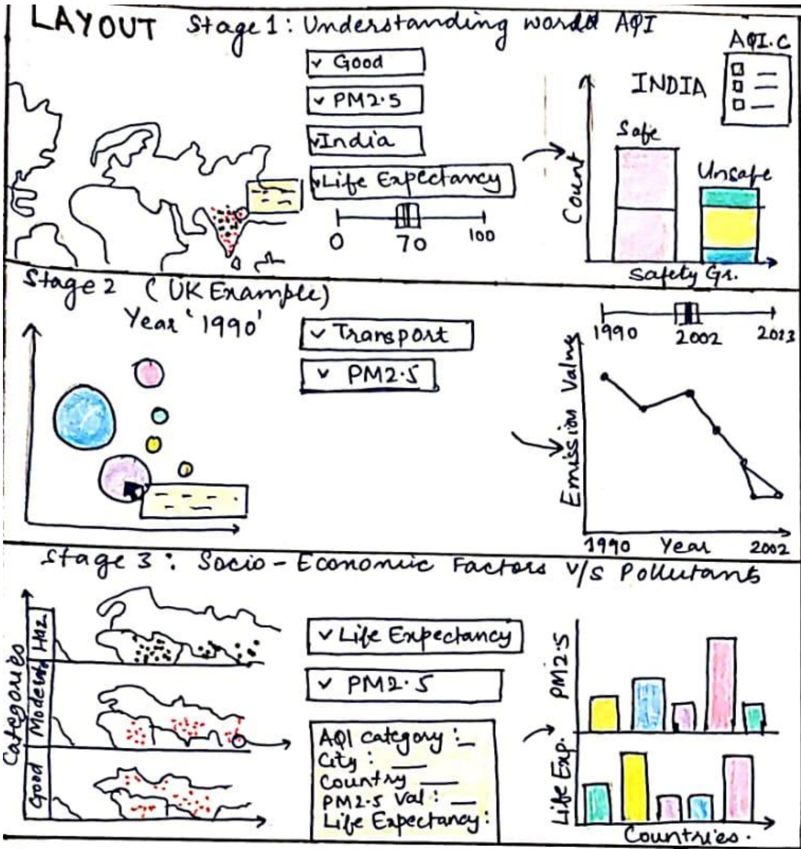
DISCUSSION

+ve : Filters are providing a better understanding Usage of the slider can help in better comparision also with the help of stacked bar plot comparision becomes easier.

-ve : If the user selects top '90' countries or a higher number the visualisation might not add enough value..

TITLE: DVP: Design
AUTHOR: Stuti Bajaj
DATE: 18-05-2025
SHEET: 5
TASK: Realisation

LAYOUT



OPERATIONS

- Various drop-down menu, along with sliders.
- Hovering over data provides insights / values.
- On clicking the data points another chart appears with seperate visualisation for better insights
- Usage of colours make charts more attractive and visible to the user.

FOCUS

- The layout of my sheet 5 represents intial designs followed by other graphs to show the relationships of AQI, socio-economic factor or trends for user to have a better understanding of the ~~data~~ data using filters, sliders, drop-down menu etc.
- The UK example helps in undesstanding the industries that are the highest or lowest contributors over time. This example can help relate with the world data.
- Air pollution may not be directly proportional to the socio-economic factors but is definitely something that matters for the country. The stage 3 helps us understand the relationships of these factors with AQI.

DISCUSSION

- A combined dataset of all 4 datasets from various sources to be used. It is cleaned using google colab by using python and wrangled.
- R shiny will be used.
- Time to build estimation 24 hrs.
- Colour choices based on Seminar material and surveys.